# Artificial Intelligence and Machine Learning
# 6CS012

**Q/A**

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**Part I – Theory Questions (Paru)**
**Long Question**:

1. You are a Machine Learning Engineer at a rapidly growing e-commerce company responsible for deploying and maintaining ML systems in production.

Identify and explain at least three real-world challenges encountered in ML systems.
Discuss consequences, solutions, and trade-offs.
Explain the importance of collaboration and modern tools.

-> In a real e-commerce business, as a Machine Learning Engineer, you will have to face various challenges when deploying and maintaining machine learning systems in production. Data drift is a common issue; it means the distribution of the data items that arrive over time are different. Consumer tastes and habits, seasonal patterns, and product demand can shift, leading to less accurate forecasts by the model. This can result in poor recommendations and low customer satisfaction if not dealt with. This problem can be addressed via monitoring systems and automated retraining pipelines, but with a high frequency of retraining, the computational cost and system complexity grows.

The other is that of latency for real-time systems like, say, a recommendation engine or search engine ranking system. High latency can impact end user experience as customers want fast response when they are shopping for products. Businesses can take advantage of distributed serving, lightweight models or model optimization to achieve lower latency. However, there is a compromise between speed and accuracy, with the optimization of the speed sometimes compromising the accuracy of the model. Model degradation and monitoring is a third big challenge. After deployment, model performance may decrease due to changing data patterns. If not monitored, these failures can be silent for many years. Model performance monitoring, continuous evaluation, and automated alerting mechanisms contribute to the reliability and tracking of model performance. But with such systems, there is a need for more infrastructure and experienced engineers.

When developing trustworthy ML systems that meet business objectives, it is crucial that data scientists, software engineers and product managers work together. In addition, automated pipelines, cloud platforms, and tools powered by LLM-enabled monitoring capabilities further enhance production-level machine learning system scalability, automation, and maintenance.

## Question 1:

- Explain techniques to reduce overfitting
- Example:
    - Dropout
    - Early stopping
    - Data augmentation
    - L2 regularization

-> One of the key issues in deep learning is overfitting, in which the model achieves high accuracy on the training set and low accuracy on an unseen test set. Several regularization methods are applied in the training of the model to avoid overfitting.

One of the effective means is dropout. This method involves randomizing the firing of some of the neurons in the process of training. This helps avoid overfitting by the neural network and enhances generalisation. It is frequently used in CNN and LSTM.

Early stopping is another useful technique. The performance of the model is validated after each epoch during training. Training will be automatically ended if the validation loss begins to rise, while the training accuracy increases. This reduces the chances of the model memorizing the training data.

The technique of data augmentation is broadly applied in image classification challenges. It creates a larger dataset by transforming images, like rotating, flipping, or zooming, them. This enables the model to capture a wider range of patterns and minimize reliance on a small amount of information. Another way of diminishing overfitting is L2 regularization. It introduces a penalty term in the loss function to penalize the model for having very large weight values. This results in a more simple and stable model. These techniques complement each other to enhance the generalisation and prediction of models.

## Question 2:

- FCN vs Autoencoder
- Architecture differences
- Purpose
- Applications

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Fully Connected Networks (FCNs) and Autoencoders are two types of neural networks, having different purposes and architectures.

The FCNs are traditional neural networks, in which neurons of each layer are connected to all neurons of the next layer. They are simple to build and have as many layers as there are input, hidden, and output layers. FCNs primarily are applied to classification and regression problems. They can be useful for tabular data, but cannot maintain spatial or sequential information. The major drawback of their implementation is that they have many parameters, and when they are used with complex data sets they can suffer from overfitting.

Unsupervised learning and dimensionality reduction are the tasks of autoencoders, however. They are designed with an encoder that compresses the input to a lower dimensional representation, and a decoder that reconstructs the original input. The primary goal is to gain knowledge of effective data representations. The applications of autoencoders include anomaly detection, image denoising and feature extraction. They are different to FCNs as they're interested in rebuilding, not predictions. They can also overfit the model if they're trained too long, though, and may learn identity mapping rather than what is useful.

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